

Hand Gesture Recognition using Surface EMG Signals: Exploratory Data Analysis of the GRABMyoPC Dataset

Introduction

Surface electromyography (sEMG) is a technique used to record the electrical activity of muscles during movement. It is widely used in applications such as hand gesture recognition, prosthetic hand control, rehabilitation and human-computer interaction.

Before building any machine learning model, it is important to understand the dataset. Therefore, Exploratory Data Analysis (EDA) was performed on the GRABMyo dataset. The analysis includes studying the dataset structure, participant information, gesture and trial distribution, missing values, duplicate records and EMG signal characteristics. Different statistical summaries and visualizations, such as histograms, boxplots, correlation heatmaps and PCA, were also used to better understand the data which will help in the next stages of the project, including data preprocessing, feature extraction and model development for hand gesture recognition.

Dataset Description

The GRABMyoPC dataset contains high-density surface EMG recordings collected from healthy participants while performing predefined hand gestures.

Dataset characteristics:

Property	Value
Sessions	3
Participants	43
Gesture Classes	17
Trials per Gesture	7

Total Recordings	15,351
Sampling Frequency	2048 Hz
EMG Channels	32

Each recording consists of synchronized EMG signals collected from multiple electrodes placed around the forearm.

Exploratory Data Analysis

The following analyses were performed:

- Dataset structure analysis
- Session distribution
- Participant analysis
- Gender distribution
- Age distribution
- Forearm length distribution
- Upper ring circumference distribution
- Wrist circumference distribution
- Gesture distribution
- Trial distribution
- Missing value analysis
- Duplicate data analysis
- EMG waveform visualization
- Signal histogram
- EMG channel boxplots
- Correlation heatmap
- Principal Component Analysis (PCA)

Key Findings

The EDA revealed several important characteristics:

- The dataset contains three recording sessions and 43 participants.
- All 17 gestures contain 903 recordings, indicating a perfectly balanced dataset.
- Each gesture was repeated seven times, resulting in equal trial distribution.

- The participant ages are primarily between 24 and 29 years, indicating a relatively homogeneous participant group.
- No missing values or duplicate EMG samples were observed.
- EMG signals fluctuate around zero, which is typical of muscle electrical activity.
- Correlation analysis indicates that neighboring EMG channels capture related muscle activation patterns.
- PCA demonstrates that the majority of signal samples share similar characteristics while preserving natural variability.

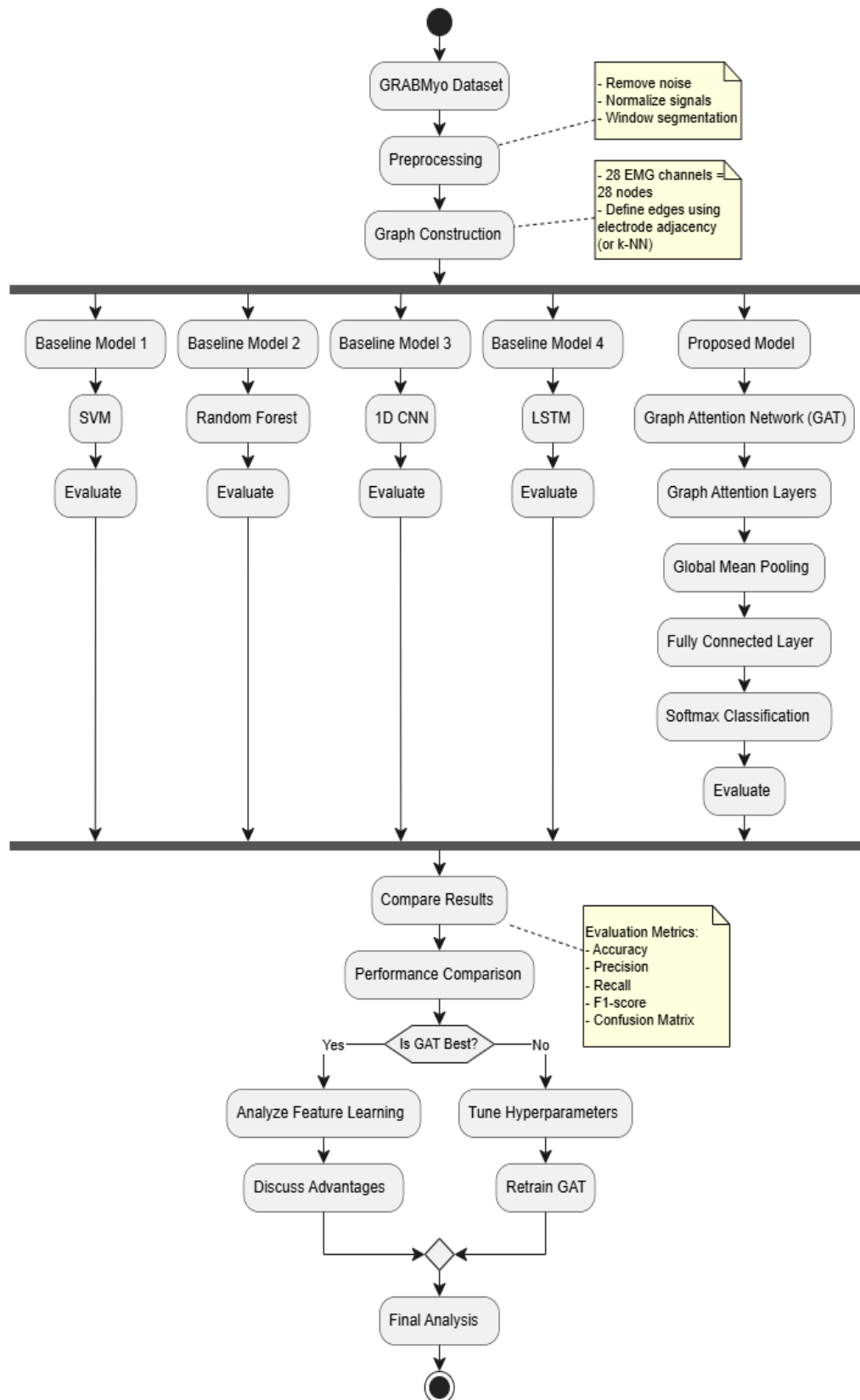
Previous Work

Reference	Task / Focus	Model / Architecture Used	Method	Accuracy / Performance	Research Gap	Proposed Solution (Paper)	Our Proposed Work	Limitation
Pradhan et al. (2022) [1]	Multi-day sEMG dataset	Dataset only (No ML model)	Longitudinal recording protocol (43 subjects, 3 sessions)	N/A	Lack of large multi-day benchmark datasets	Introduced the GRAB Myo dataset	Use the full GRAB Myo dataset to train and compare multiple deep learning models for gesture recognition.	Dataset quality depends on consistent electrode placement.

Kok et al. (2024) [2]	Prosthetic control	Support Vector Machine (SVM) (compared with KNN & Naïve Bayes)	DWT + composite time-domain feature extraction	SVM outperformed KNN and NB	Hand-crafted features are sensitive to electrode shifts	Feature engineering and feature reduction	Replace manual feature engineering with automatic deep feature learning using CNN-based models.	Requires manual feature extraction.
Bilgin et al. (2023) [3]	Biometric classification	CNN, LSTM	Empirical Modal Decomposition (EMD) + IMF statistical features	CNN: 95.57%, LSTM: 93.88%	Evaluated only 10 subjects performing one gesture	Combined EMD with deep learning	Train on all GRABM yo gestures and all subjects for more generalized recognition.	Small dataset and single-gesture evaluation.
Aarotale & Rattani (2024) [4]	Hand Gesture Recognition	1D Dilated CNN	Fused time-domain descriptors + dilated convolutions	97.00%	Session-to-session variability still affects performance	Improved receptive field using dilated CNN	Investigate more robust architectures and evaluate generalization across multiple recording sessions.	No real-time or hardware validation.

Cansız et al. (2025) [5]	Gesture + Subject Identification	Hierarchical Cascade Deep Learning Network	Two-stage cascade (gesture → subject)	71.62% Joint Average	Error propagation between cascade stages	Separate gesture recognition and biometric identification	Develop a single end-to-end gesture recognition model without cascading to reduce accumulated errors.	Performance depends on the first-stage prediction.
Kim et al. (2025) [6]	Biometric Authentication	Convolutional Siamese Network	N-pair loss, shift-equivariant padding, augmentation	EER: 5.62%, 4.92%	Focuses on authentication instead of gesture recognition	Robust cross-day biometric learning	Adapt modern deep learning architectures specifically for multi-class hand gesture recognition rather than authentication.	Limited to biometric verification tasks.

Proposed Work



Conclusion

The exploratory data analysis confirms that the GRABMyoPC dataset is well organized, balanced and suitable for gesture recognition research as no missing or duplicate signal samples. Also, participant characteristics are consistent and the EMG signals exhibit expected physiological behavior, indicating a strong foundation for developing and evaluating machine learning and deep learning models for hand gesture recognition.

References

1. Pradhan, Ashirbad, He, Jiayuan, and Jiang, Ning (2022). Multi-day dataset of forearm and wrist electromyogram for hand gesture recognition and biometrics. *Scientific Data*, 9(1). <https://doi.org/10.1038/s41597-022-01836-y>
2. Kok, Chiang Liang et al. (2024). Machine Learning-Based Feature Extraction and Classification of EMG Signals for Intuitive Prosthetic Control. *Applied Sciences*, 14(13), 5784. <https://doi.org/10.3390/app14135784>
3. BİLGİN, Bekir, GÜRSOY, Mehmet İsmail, and ALKAN, Ahmet (2023). Biometric Personal Classification with Deep Learning Using EMG Signals. *Bilge International Journal of Science and Technology Research*, 7(2), 156-161. <https://doi.org/10.30516/bilgesci.1344337>
4. Pradhan, A., He, J. & Jiang, N. Multi-day dataset of forearm and wrist electromyogram for hand gesture recognition and biometrics. *Sci Data* **9**, 733 (2022). <https://doi.org/10.1038/s41597-022-01836-y>
5. Cansiz, Berke et al. (2025). Hierarchical Cascade Deep Learning for EMG-Based Behavioral Biometrics: Gesture and Subject Classification. *IEEE Access*, 13, 124115-124128. <https://doi.org/10.1109/access.2025.3588791>
6. Kim, Youngsam, Roh, Jong-hyuk, and Kim, Soohyung (2025). Siamese network-based user-independent model for surface electromyogram biometric authentication. *ETRI Journal*, 47(6), 1163-1177. <https://doi.org/10.4218/etrij.2024-0370>

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